

computational game theory

com810s

repeated games & learning dynamics

intro

- repeat a game T times and discover more behaviors by players
- finite and infinite repetition

finitely repeated games

stage game

- P : finite set of n **players**
- A_i : set of **actions** available to player i

$$\mathcal{G} = \langle P, A, u \rangle$$

$$A = A_1 \times A_2 \times \dots \times A_n$$

$$u = (u_1, u_2, \dots, u_n)$$

$$u_l : A_l \rightarrow \mathbb{R}$$

- T: number of iterations
- payoffs represent the sum of payoffs up to the following iteration

$$\mathcal{G}(\mathbf{T})$$

$$\mathbf{T} = [k]$$

- history
 - sequence of plays during iterations
- strategy
 - plan of actions at each step of repeated games
- equilibrium path
 - sequence of outcomes at each stage game

extensive-form game

$$\mathcal{G} = \langle P, A, T, B, u \rangle$$

- find the PSNE for the stage game

$$A = A_1 \times A_2$$

- play the game for two periods and find cumulative payoffs

$$A_1 = \{a_1, a_2, a_3\}$$

$$A_2 = \{b_1, b_2, b_3\}$$

$$M = \begin{bmatrix} 8 & 0 & 1 \\ 0 & 5 & 0 \\ 9 & 0 & 3 \end{bmatrix}; \quad N = \begin{bmatrix} 8 & 0 & 9 \\ 0 & 5 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

- the two PSNE are $\{a_2, b_2\}$ and $\{a_3, b_3\}$
- player 1:
 - play a_1 at iteration 1; then if player 2 played b_1 at iteration 1, play a_2 , else play a_3
- player 2:
 - play b_1 at iteration 1; then if player 1 played a_1 at iteration 1, play b_2 , else play b_3
- payoff for each player is 13
- note: these strategies can yield a SPNE (homework: prove it)

$$\mathcal{G} = \langle P, A, T, B, u \rangle$$

$$A = A_1 \times A_2$$

$$A_1 = \{a_1, a_2\}$$

$$A_2 = \{b_1, b_2\}$$

$$M = \begin{bmatrix} 5 & 0 \\ 6 & 1 \end{bmatrix}; \quad N = \begin{bmatrix} 5 & 6 \\ 0 & 1 \end{bmatrix}$$

- find the PSNE for the stage game
- represent the repeated game in its extensive-form for two iterations
 - are there subgames?
 - what about SPNE?
 - does the equilibrium change?

$$\mathcal{G} = \langle P, A, T, B, u \rangle$$

- what happens when the game is updated to

$$A = A_1 \times A_2$$

- explain the increment

$$A_1 = \{a_1, a_2\}$$

- can we generalise to n iterations?

$$A_2 = \{b_1, b_2\}$$

$$M = \begin{bmatrix} 6 & 1 \\ 7 & 2 \end{bmatrix}; \quad N = \begin{bmatrix} 6 & 7 \\ 1 & 2 \end{bmatrix}$$

- summary
 - if the stage game has a unique subgame perfect equilibrium, then the repeated game has a unique subgame perfect equilibrium
 - when there are multiple equilibria, not all outcome paths can be a result of a subgame perfect equilibrium
 - a large number of outcome paths can result in a subgame perfect equilibrium

infinitely repeated games

- discounted payoffs
 - use a discount factor for future payoffs (present value)

$$\delta \in [0, 1]$$

$$V_{i,j}^t = u_{i,j}^0 + \delta u_{i,j}^1 + \dots + \delta^t u_{i,j}^t$$

$$\sum_{t=\tau}^{\infty} \delta^t u = \delta^{\tau} \frac{u}{1 - \delta}$$

- single-deviation principle
 - a strategy profile s is an SPNE iff there is no profitable single-deviation for any player in any subgame

- example of deviation
 - play a strategy up to step $t - 1$
 - deviate at step t
 - return to the old strategy at $t + 1$

- augmented/reduced stage game

$$u_l(a \mid s^*, h) = u_l(a) + \delta \times v_l^{t+1}$$

generic strategies

- grim-trigger
 - play a cooperative strategy until a player deviates, then deviate forever

$$\mathcal{G} = \langle P, A, T, B, u \rangle$$

- consider the rhs game

$$A_1 = \{a_1, a_2\}$$

$$A_2 = \{b_1, b_2\}$$

$$M = \begin{bmatrix} 5 & 0 \\ 6 & 1 \end{bmatrix}; \quad N = \begin{bmatrix} 5 & 6 \\ 0 & 1 \end{bmatrix}$$

- scenario 1
 - only a_1, b_1 strategy profile
- scenario 2
 - deviation from scenario 1

$$V_1 = 5 + 5\delta + 5\delta^2 + \dots$$

$$V_2 = 1 + \delta + \delta^2 + \dots$$

- no deviation scenario
- solve case where a_1, b_1 is a NE

$$M = \begin{bmatrix} 5 + \delta V_1 & 0 + \delta V_2 \\ 6 + \delta V_2 & 1 + \delta V_2 \end{bmatrix}; \quad N = \begin{bmatrix} 5 + \delta V_1 & 6 + \delta V_2 \\ 0 + \delta V_2 & 1 + \delta V_2 \end{bmatrix}$$

- deviation scenario
- solve case where a_2, b_2 is a NE
 - a_2, b_2 is a NE

$$M = \begin{bmatrix} 5 + \delta V_2 & 0 + \delta V_2 \\ 6 + \delta V_2 & 1 + \delta V_2 \end{bmatrix}; \quad N = \begin{bmatrix} 5 + \delta V_2 & 6 + \delta V_2 \\ 0 + \delta V_2 & 1 + \delta V_2 \end{bmatrix}$$

- conclusion
 - condition for both scenarios (histories) to meet the single-deviation principle and therefore yield SPNE

- Tit-for-tat
 - deviate in the next iteration only if the other player deviates in the current iteration

- assess the following scenarios
 - $\{a_1, b_1\}$ is played at step t
 - cooperative scenario repeated throughout
 - $\{a_1, b_2\}$ or $\{a_2, b_1\}$ is played at step t
 - one-sided deviation repeated throughout
 - $\{a_2, b_2\}$ is played at t
 - deviation for the remaining iterations

$$5 + 5\delta + 5\delta^2 + \dots = \frac{5}{1 - \delta}$$

$$6 + 0\delta + 6\delta^2 + \dots = \frac{6}{1 - \delta^2}$$

$$0 + 6\delta + 0\delta^2 + \dots = \frac{6\delta}{1 - \delta^2}$$

$$1 + \delta + \delta^2 + \dots = \frac{\delta}{1 - \delta}$$

$$\mathbf{M} = \begin{bmatrix} 5 + \delta \frac{5}{1 - \delta} & 0 + \delta \frac{6}{1 - \delta^2} \\ 6 + \delta \frac{6}{1 - \delta^2} & 1 + \frac{\delta}{1 - \delta} \end{bmatrix}; \quad \mathbf{N} = \begin{bmatrix} 5 + \delta \frac{5}{1 - \delta} & 6 + \delta \frac{6}{1 - \delta^2} \\ 0 + \delta \frac{6}{1 - \delta^2} & 1 + \frac{\delta}{1 - \delta} \end{bmatrix}$$

- analysis of NE for some scenarios (a_1b_1 , a_1b_2) with single-deviation leads to a contradiction except for equality
 - discuss the value of discount
- in conclusion tit-for-tat does not yield a SPNE

learning in games

fictitious play

- belief-based learning rule
 - players form belief from other players' moves and act rationally according to that belief

- action observation
 - number of times player j observed actions from other players before time t
- player j assumes that other players uses a mixed strategy and updates its belief about the strategy at each step
- other player's strategy follows empirical distribution of past play

$$\eta_l^t : \mathbf{S}_{-l} \rightarrow \mathbb{N}$$

$$\mu_l^t = \frac{\eta_l^t(a_{-l})}{\sum_{\hat{a}_j} \eta_l(\hat{a}_j)}$$

- player j chooses action to maximize its payoff

$$a_i^t \in \max u_i(a_i, \mu_i^t)$$

$$\mathcal{G} = \langle P, A, T, B, u \rangle$$

$$A_1 = \{a_1, a_2\}; \quad A_2 = \{b_1, b_2\}$$

- example
 - two players (P_1 and P_2)
 - two actions per player
 - payoffs as indicated in the matrices

$$M = \begin{bmatrix} 3 & 0 \\ 4 & 1 \end{bmatrix}; \quad N = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\eta_1^0 = [3, 0] \quad \eta_2^0 = [1, 2.5]$$

- algorithm
 - update count and belief
 - compute payoffs and choose action
 - repeat...

- algorithm converges to equilibrium $\{a_2b_2\}$
- can we prove it?

- caution!
 - convergence is not always guaranteed

regret matching

- use regret/gain instead of count

$$\mathcal{G} = \langle P, A, T, B, u \rangle$$

$$A_1 = \{a_1, a_2\}; \quad A_2 = \{b_1, b_2\}$$

- example
 - two players (P_1 and P_2)
 - two actions per player
 - payoffs as indicated in the matrices

$$M = \begin{bmatrix} 3 & 0 \\ 4 & 1 \end{bmatrix}; \quad N = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

$$R_1^0 = [1, 1] \quad R_2^0 = [1, 1]$$

- algorithm
 - update belief
 - compute payoff and expected payoff of mixture
 - define gain
 - only consider surplus or zero
 - update regret vector with gain
 - repeat...

- reach equilibrium when all gains are 0

summary

- repeated games
 - finite iterations
 - infinite iterations
- learning to find an equilibrium
 - fictitious play
 - regret matching